

Whose Seat Am I In?—Solutions

Spring 2020 ARML Power Contest

Problem 1. Here are the eight possible seating arrangements from four people, with just the first being absent-minded.

Person 1 sits in seat	with probability	Person 2 then sits in seat	with probability	Person 3 then sits in seat	with probability	Person 4 must sit in seat	Notation	Overall probability
1	1/4	2	1	3	1	4	(1, 2, 3, 4)	1/4
2	1/4	1	1/3	3	1	4	(2, 1, 3, 4)	1/12
		3	1/3	1	1/2	4	(3, 1, 2, 4)	1/24
		3	1/3	4	1/2	1	(4, 1, 2, 3)	1/24
		4	1/3	3	1	1	(4, 1, 3, 2)	1/12
3	1/4	2	1	1	1/2	4	(3, 2, 1, 4)	1/8
		2	1	4	1/2	1	(4, 2, 1, 3)	1/8
4	1/4	2	1	3	1	1	(4, 2, 3, 1)	1/4

Just to double-check: the sum of all the probabilities in the last column does add to one, and the probabilities where person 4 is in seat 4 (rows 1, 2, 3, and 6 in the table) do add to 1/2 as advertised.

Problem 2. From the table above, note that the 4 is in either the first entry or the last entry in each quadruple, so person 4 must be in either seat 1 or seat 4.

Problem 3. Consider person k and seat j with $1 \neq j < k$. Person k cannot sit in seat j because if person 1 didn't accidentally sit in seat j , then person j came into the room before person k and her seat was available, so she sat in it. In either case, seat j is now unavailable for person k to sit in.

Problem 4. Let the subset be $\{1, p_2, p_3, \dots, p_k\}$ where $1 < p_2 < p_3 < \dots < p_k$. The only way in which these people, and no others can sit in the wrong seats is if person 1 sits in seat p_2 , person p_2 sits in seat p_3 , and so on, with the last person in the subset, person p_k , sitting in seat 1. Clearly for only these people to be in the wrong seats and everyone else to be in their assigned seats, none of these people can sit in a seat other than the seats with numbers in this subset, because that would displace someone whose number is not in the subset. And by the result of the previous problem everyone in this subset must either be in his or her own seat, a higher-numbered seat, or seat 1. None can be in their own seat if they are all to be in a wrong seat. So person p_k must be in seat 1 because there is no wrong-numbered seat with number higher than p_k . Working backward, since seat 1 is now filled by person p_k , the only wrong seat available for person p_{k-1} is seat p_k . Then since person p_{k-2} can't be in seat 1 or seat p_k he or she must be in seat p_{k-1} , and so on until person 1 must be found in seat p_2 . A full proof would use induction, but the problem only asked for an explanation rather than a complete proof.

Problem 5. Two approaches suggest themselves.

The first is to build on the previous problem. Clearly there is one arrangement where everyone is in the correct seat, and no arrangement where exactly one person is in the wrong seat. Now if person 1 sits in his correct seat 1, so will everyone else, so if there is anyone in a wrong seat, person 1 must be among them. We know from the previous problem that all sets of at least two people that include person 1 give rise to exactly one seating arrangement where that set is the set of people in wrong seats. Now a subset of two or more people that includes person 1 can be realized as any non-empty subset of $\{2, 3, \dots, n\}$ with person 1 added to it. Since there are $n - 1$ elements in $\{2, 3, \dots, n\}$, it is well-known that there are 2^{n-1} subsets, and hence $2^{n-1} - 1$ non-empty subsets. Together with the one arrangement where everyone is seated correctly, the total is 2^{n-1} .

A second approach is to condition on whether or not the last person is sitting in her correct seat. Let the number of seating arrangements with n people be denoted A_n . If person n is in the correct seat, then each arrangement of $n - 1$ people becomes an arrangement of n people by placing person n in her correct seat. So there are A_{n-1} arrangements of n -people where person n is in the correct seat. Each arrangement of $n - 1$ people where some people are in the wrong seat also becomes an arrangement of n people with some in wrong seats by putting person n in seat 1 and the person who had sat in seat 1 into seat n (this is certainly an arrangement with some people in wrong seats, and by the previous problem there is only one such arrangement possible for any particular subset of people). This adds $A_{n-1} - 1$ more arrangements (because some of the first $n - 1$ people had to be in incorrect seats, so the arrangement of $n - 1$ people where everyone is in his or her correct seat has been discounted). Finally, there is one more arrangement where person 1 and person n swap seats, so the total is $A_n = A_{n-1} + (A_{n-1} - 1) + 1 = 2A_{n-1}$. Together with the obvious initial values $A_1 = 1$ and $A_2 = 2$, and the values $A_3 = 4$ and $A_4 = 8$ as have been discovered in earlier work, this recursion clearly results in $A_n = 2^{n-1}$.

Problem 6. It is tempting to observe that, in the second solution to the previous problem, half of the arrangements had person n in her correct seat and half did not and conclude that the probability is $1/2$. But that conclusion is not justified. Not all arrangements have the same probability, and this must be taken into account. A nice recursive argument is as follows. It has already been observed that for $n = 2$ people the probability of person n sitting in her correct seat is $1/2$.

Using the principal of strong induction, assume it has already been shown that the desired probability is $1/2$ when there are between 2 and n people, inclusive. Consider the case of $n + 1$ people. There is a $1/(n + 1)$ probability that person 1 will sit in seat 1, in which case everyone will sit in his or her own seat, including person $n + 1$. There is also a $1/(n + 1)$ probability that person 1 will sit in seat $n + 1$, so that person $n + 1$ will now definitely not be in her own seat. That leaves a probability of $\frac{n-1}{n+1}$ that person 1 will sit in seat k , with $2 \leq k \leq n$. Now all people with numbers in the range $2-k-1$ (if any) will sit in their own seats. So seats $2-k$ are filled by people $1-k-1$. Person k now must sit in a seat at random, and the situation is as if there were $(n + 1) - (k - 1) = n + 2 - k$ people waiting for seats, with person k taking the role of the absent-minded person 1. Since $2 \leq k \leq n$, $n + 2 - k$ is between 2 and n , so the probability that the last person sits in her own seat has inductively been assumed to be $1/2$. Thus, the total probability that person $n + 1$ is in her own seat is

$$\frac{1}{n+1} \cdot 1 + \frac{1}{n+1} \cdot 0 + \frac{n-1}{n+1} \cdot \frac{1}{2} = \frac{1}{2}$$

as desired.

Problem 7. These probabilities can be computed from the tables for $n = 3$ and $n = 4$ that have already been created.

- (a) Revisiting the table in the beginning of the question, person 2 is in her correct seat unless person 1 sat there first, which has probability $1/3$, so the probability that person 2 is in her correct seat is $\boxed{2/3}$.
- (b) Examining the table from problem 1 shows person 3 to be sitting in her correct seat in rows 1, 2, 5, and 8, whose probabilities sum to $1/4 + 1/12 + 1/12 + 1/4 = \boxed{2/3}$.
- (c) Again referring back to the table from problem 1, person 2 is in her correct seat unless person 1 already sat there (which happens $1/4$ of the time) so her probability of being in the correct seat is $\boxed{3/4}$.

Problem 8. The problem is similar to problem 6. In that problem, it was shown that (if there are at least two people) than the last person sits in her own seat with probability $1/2$. This problem extends that to say that if there are at least three people, then person $n - 1$ sits in her own seat with probability $2/3$; if there are at least four people than person $n - 2$ sits in her own seat with probability $3/4$ and so forth. The probabilities are determined by how far the person is from the *end* of the line.

Let $j = n - k$ denote how far the person is from the end of the line. Thus, $j = 0$ is the last person in line (whom we already know has probability $1/2$ to sit in her own seat), $j = 1$ represents the second-to-last person, and so on. The goal is to prove that person $n - j$ sits in her own seat with probability $\frac{j+1}{j+2}$.

For a fixed j , the base case occurs when $k = n - j = 2$. Person 2 always sits in her own seat with probability $\frac{n-1}{n}$ because there is only a $\frac{1}{n}$ probability that person 1 sits in seat 2, and otherwise person 2 gets to sit in her own seat. So for all choices of j , with $k = 2$, the probability is $\frac{n-1}{n}$. Substituting $n = k + j$ and $k = 2$ leaves a probability of $\frac{j+1}{j+2}$ as desired.

Now using strong induction, assume that the probability that person $n - j$ is in her own seat is $\frac{j+1}{j+2}$ for all values of k from 2 up to $n - j$. Now let $k = n + 1 - j$. There are four possibilities:

- Person 1 sits in one of the seats numbered at least $k + 1$. Then person k will sit in her own seat. There are j of these seats, so this happens with probability $\frac{j}{n+1}$.
- Person 1 sits in seat k . Then person k will not get to sit in her own seat. This happens with probability $\frac{1}{n+1}$.
- Person 1 sits in seat 1, and everyone gets to sit in his or her own seat. This happens with probability $\frac{1}{n+1}$.
- Person 1 sits in a seat s where s is numbered from 2 to $k - 1$. Just like problem 6, if person s is renamed person 1 and people $1-s-1$ and seats $2-s$ are ignored, by the inductive assumption person k now gets to sit in her seat with probability $\frac{j+1}{j+2}$. Since there are $k - 2$ seats in this range, this scenario happens with probability $\frac{k-2}{n+1}$.

Putting these pieces together, the probability that person k is sitting in her own seat is

$$\begin{aligned} & \frac{j}{n+1} \cdot 1 + \frac{1}{n+1} \cdot 0 + \frac{1}{n+1} \cdot 1 + \frac{k-2}{n+1} \frac{j+1}{j+2} \\ &= \frac{1}{n+1} \left(j + 1 + \frac{(k-2)(j+1)}{j+2} \right) \\ &= \frac{1}{n+1} \frac{(j+1)(j+2+k-2)}{j+2} \end{aligned}$$

Noting that $k = n + 1 - j$, this simplifies to $\frac{j+1}{j+2}$ as desired. Since the value of j was arbitrary and induction has now shown that the result holds for all values of k (and hence $n = j + k$) the proof is complete.

Alternate Solution The following is a composite of several student solutions that are a lot more elegant than the above computation. From problem 3 person k cannot sit in seats $2-k-1$. So there are $k-2$ seats in which she never sits, leaving $n-k+2$ seats in which person k is allowed to sit. She will sit in her correct seat *unless* the first person ahead of her that chose a seat numbered either 1 or $k-n$ specifically chose seat k . Since this choice is random and equally likely to choose any seat, the chance that seat k is chosen is $1/(n-k+2)$ leaving a probability of $\frac{n-k+1}{n-k+2}$ that seat k is available.

Problem 9.

(a) Running through all possible values of s :

$s = 0$ Consulting the table shows one case where no one sits in the correct seat, with probability $1/6$.

$s = 1$ There are two arrangements in the table with just one person in the correct seat; their total probability is $1/2$.

$s = 2$ This is impossible.

$s = 3$ There is only one arrangement where everyone sits in the correct seat, and it has probability $1/3$.

(b) The expected value of the number is the sum of the number of people in correct seats times the probability that this number of people are in their correct seats. In this case,

$$\sum_{s=0}^3 s \cdot p(s) = 0 \cdot \frac{1}{6} + 1 \cdot \frac{1}{2} + 2 \cdot 0 + 3 \cdot \frac{1}{3} = \boxed{\frac{3}{2}}.$$

Problem 10.

Repeating the above calculations using the table in problem 1:

$s = 0$ One arrangement with probability $1/24$.

$s = 1$ Three arrangements with total probability $1/4$.

$s = 2$ Three arrangements with total probability $11/24$.

$s = 3$ Not possible.

$s = 4$ One arrangement with probability $1/4$.

(a) This time the total is

$$0 \cdot \frac{1}{24} + 1 \cdot \frac{1}{4} + 2 \cdot \frac{11}{24} + 3 \cdot 0 + 4 \cdot \frac{1}{4} = \boxed{\frac{13}{6}}$$

Problem 11. If the expected value of the number of people in correct seats is denoted E_n ,

then either the explicit formula $\boxed{n - \sum_{k=1}^{n-1} \frac{1}{k} = n - H_{n-1}}$ (where H_m is the m^{th} harmonic number,

$H_m = 1 + 1/2 + 1/3 + \cdots + 1/m$) or the recursive formula $\boxed{E_n = \begin{cases} 1 & n = 1 \\ E_{n-1} + \frac{n-1}{n} & n > 1 \end{cases}}$

(or equivalents) are acceptable.

These formulas can be arrived at as follows. Clearly $E_1 = 1$ and it is fairly straightforward to compute $E_2 = 1$. The previous problems computed $E_3 = 3/2$ and $E_4 = 13/6$. Checking the differences between these, they grow by 0, then $1/2$, then $2/3$ so the recursive formula might be conjectured based on these samples. If the recursive formula is correct, it is straightforward to obtain the explicit formula since $E_n = E_{n-1} + \frac{n-2}{n-1} = E_{n-1} + 1 - \frac{1}{n-1} = E_{n-2} + 2 - \frac{1}{n-1} - \frac{1}{n-2} = \dots$ until $E_1 = 1$ is reached.

Now to justify the recursion! Claim:

$$E_{n+1} = \frac{1}{n+1} \left(n+1 + \sum_{k=0}^{n-1} \left(k + E_{n-k} - \frac{1}{n-k} \right) \right)$$

This follows by examining the possible seats where person 1 might sit. Each seat has probability $\frac{1}{n+1}$, explaining the fraction in front of the expression on the right-hand side. If person 1 sits in seat 1, then all $n+1$ people will sit in their correct seats, which is the $n+1$ term outside the sum on the right-hand side.

What is the expected number of people in correct seats if person 1 sits in one of the seats from 2 up to $n+1$? Assume he sits in seat $k+2$ (so that k ranges from 0 to $n-1$). Then persons 2 through $k+1$ (a total of k people) sit in their correct seats before person $k+2$ walks into the room and notices her seat is taken. That accounts for the k in the sum.

When person $k+2$ walks into the room, she sees seat 1 empty, seats 2 through $k+1$ (correctly) filled, and her seat $k+2$ incorrectly filled. The situation is just like the regular game played with $n-k$ people *except* that she can't sit in her own seat—she would have to sit in seat 1 instead. Normally if she sits in her own seat so will everyone else, while if she sits in someone else's seat, a chain of people will be displaced until someone else randomly sits in her seat. But in this case the chain will end when someone sits in seat 1. Either in the regular game or in this case, everyone in the chain is in the wrong seat and everyone not in the chain is in the right seat, so this doesn't affect the expected value. But in the case where person $k+2$ was supposed to sit in her own seat and instead had to sit in seat 1, only the remaining $n-k-1$ people sit in their own seats, not $n-k$ people because person $k+2$ is in seat 1 instead of her own seat. Thus, in computing the expected value for this situation, the number of people in correct seats is reduced by one in the case where everyone is supposed to sit in their own seat, which happens with probability $\frac{1}{n-k}$. So instead of E_{n-k} the expected number of additional people sitting in correct seats once person $k+2$ walks into the room is $E_{n-k} - \frac{1}{n-k}$. This accounts for the remainder of the terms in the sum.

Finally, the sum is manipulated into the recursion formula:

$$\begin{aligned} E_{n+1} &= \frac{1}{n+1} \left(n+1 + \sum_{k=0}^{n-1} \left(k + E_{n-k} - \frac{1}{n-k} \right) \right) \\ &= \frac{1}{n+1} \left(n+1 + E_n - \frac{1}{n} + \sum_{k=1}^{n-1} \left(k + E_{n-k} - \frac{1}{n-k} \right) \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n+1} \left(E_n + \frac{n-1}{n} + n + \sum_{k=0}^{n-2} \left(k+1 + E_{n-1-k} - \frac{1}{n-1-k} \right) \right) \\
&= \frac{1}{n+1} \left(E_n + \frac{n-1}{n} + n + n-1 + \sum_{k=0}^{n-2} \left(k + E_{n-1-k} - \frac{1}{n-1-k} \right) \right) \\
&= \frac{1}{n+1} \left(E_n + \frac{n-1}{n} + \frac{n(n-1)}{n} + \left[n + \sum_{k=0}^{n-2} \left(k + E_{n-1-k} - \frac{1}{n-1-k} \right) \right] \right) \\
&= \frac{1}{n+1} \left(E_n + \frac{n^2-1}{n} + nE_n \right) \\
&= \frac{1}{n+1} \left((n+1)E_n + \frac{(n+1)(n-1)}{n} \right) \\
&= E_n + \frac{n-1}{n}.
\end{aligned}$$

Alternate Solution Many student solutions noted a much simpler approach. The probability that person 1 is in his own seat is $1/n$ while by problem 8 the probability for person k to be in her own seat, $k > 1$, is $\frac{n-k+1}{n-k+2}$. The result now follows from linearity of expectation.

Problem 12.

(a) Since person 1 has three choices and person 2 has two choices of where to sit, and then person 3 sits in whatever seat is still available, there are $3 \cdot 2 \cdot 1 = \boxed{6}$ possible seating arrangements.

(b) Of these six arrangements, each of which has equal probability of occurring, person 3 is in her seat for two of them, $(1, 2, 3)$ and $(2, 1, 3)$, leading to a probability of $2/6 = \boxed{1/3}$.

Problem 13.

(a) If person 1 sits in seat 1 the remaining three people are in the $n = 3, a = 1$ case where there are four arrangements. Similarly if person 1 sits in seat 2 there are four arrangements possible. If person 1 sits in seat 3 then wherever person 2 sits (three choices), person 3 will have two choices, making six more arrangements. If person 1 sits in seat 4, there are four possibilities: $(2, 4, 3, 1)$, $(4, 2, 3, 1)$, $(3, 4, 2, 1)$, and $(4, 3, 2, 1)$. There are a total of $\boxed{18}$ possibilities.

(b) Person 3 gets to sit in her own seat unless person 1 or person 2 sat there. The probability is $3/4$ that person 1 did not sit there, and then a probability of $2/3$ that person 2 didn't sit there either, making the probability $\frac{3}{4} \cdot \frac{2}{3} = \frac{2}{4} = \frac{1}{2}$ that neither sat in seat 3, so a complementary probability of $\boxed{\frac{1}{2}}$ that person 3 can sit there.

(c) The arrangements where person 4 is in seat 4 are $(1, 2, 3, 4)$, $(2, 1, 3, 4)$, $(1, 3, 2, 4)$, $(3, 1, 2, 4)$, $(2, 3, 1, 4)$, and $(3, 2, 1, 4)$. These occur with probabilities $1/12$, $1/12$, $1/24$, $1/24$, $1/24$, and $1/24$ respectively, which add to $\boxed{\frac{1}{3}}$.

Problem 14.

(a) There are $\binom{5}{2} = 10$ possibilities where persons 1 and 2 could sit. Count based on these possibilities, then double the result (because persons 1 and 2 could swap places). If persons 1 and 2 are in seats:

- 1 and 2: everyone else sits in her correct seat—**1** arrangement.
- 1 and 3: the remaining people are in the $n = 3, a = 1$ case (reassigning person 3 to sit in seat 2), which has **4** arrangements.
- 1 and 4: person 3 sits in her correct seat, leaving **2** choices for the others.
- 1 and 5: since persons 3 and 4 get their correct seat, this leads to only **1** arrangement.
- 2 and 3: like the case of 1 and 3, there are **4** arrangements.
- 2 and 4: like the case of 1 and 4, there are **2** arrangements.
- 2 and 5: like the case of 1 and 5, there is **1** arrangement.
- 3 and 4: the remaining people all sit randomly, so **6** arrangements.
- 3 and 5: if person 3 sits in seat 1 or seat 2, persons 4 and 5 will have no choices in their seats. But if person 3 sits in seat 4, person 4 will then choose between two seats. Altogether this gives **4** arrangements.
- 4 and 5: person 3 will sit in her own seat, and then person 4 will choose between the **2** remaining seats.

Adding the numbers in bold leads to 27 arrangements, which is doubled (see above) to yield a final total of $\boxed{54}$ seating arrangements.

(b) This is actually a little easier to count! Since the first three people choose randomly where they will sit, the only thing that matters is whether or not one of them sits in seat 4. If one does, then person 4 will choose one of the remaining two seats randomly. But if seat 4 is open, person 4 sits there and person 5 takes whatever seat is left. Of the $\binom{5}{3} = 10$ choices of where the first three people could sit, four of them miss seat 4 and six do not. So there are $6 \cdot 2 + 4 = 16$ possibilities for where persons 4 and 5 are sitting. For each of these, the first three people sit in their three seats in a random order, so six permutations. Thus, there are a total of $6 \cdot 16 = \boxed{96}$ possible seating arrangements.

Problem 15. Once person k enters the room, all lower-numbered people have already entered. Each person with a number lower than k and higher than a will sit in her own seat unless it is already taken. Thus all seats with numbers lower than k and higher than a are already taken by the time person k enters, meaning she must sit in a seat with a number either at least k or at most a .

Problem 16. Since part (a) is the special case of part (b) where $k = n$, only part (b) will be proven.

The strategy is similar to problem 8, where what matters is the person's position relative to the end of the line, not the beginning. So again let $j = n - k$ be the person's distance from the end of the line. Since $k = n - j > a$ we have $n > a + j$. So for any given j , the minimum possible number of people waiting to get into the room is $a + j + 1$. Let $c = n - a$, the number of clear-headed (i.e., not absent-minded) people. Note that c must be at least one since $c + a = n \geq k > a$.

Denote by $P(a, c, j)$ the probability that, given a absent-minded people followed by c clear-headed people, that person $k = n - j = a + c - j$ will sit in her own seat. The goal will be to prove that $P(a, c, j) = \frac{j+1}{j+a+1}$. Substituting $j = n - k$ then gives the desired result.

It should be clear that when $a = 0$, $P(0, c, j) = 1 = \frac{j+1}{j+a+1}$ since if there are no absent-minded people then everyone will sit in their assigned seat. It should also be clear that $j < c$ for $j = n - k = a + c - k$ and $k > a$. In fact, when $j = c - 1$ then $k - a = 1$ and person k is going to choose a seat immediately after all the absent-minded people choose their seats. In this case, a seats are taken at random, so the probability that person k gets her own seat is $\frac{n-a}{n}$. Thus, $P(a, c, c-1) = \frac{n-a}{n}$. Note that this equals $\frac{a+c-a}{a+c} = \frac{c}{a+c} = \frac{j+1}{j+a+1}$ as desired.

To compute $P(a, c, j)$, let person 1 randomly choose to sit in seat s .

With probability $\frac{a}{n}$, s will be less than or equal to a . Swap the names on seats 1 and s . So now person 1 is in the correct seat, there are $a - 1$ more absent-minded people to sit in $n - 1$ seats, so the probability that person k now gets to sit in her own seat is $P(a - 1, c, j)$.

With probability $\frac{k-1-a}{n}$, s will be larger than a but less than k . In this case, someone who wasn't absent-minded will become displaced from their seat, and that person will choose a random seat before person k gets to look for a seat. So essentially one seat has been taken away, but there are the same number of absent-minded people ahead of person k . The probability that person k will get her own seat in this case is $P(a, c - 1, j)$.

With probability $\frac{1}{n}$, s will equal k . Then person k cannot sit in her own seat, and this case contributes zero to the overall probability.

Finally, with probability $\frac{j}{n}$, s will be greater than k . Person 1 and seat s are now eliminated leaving $n - 1$ people of whom $a - 1$ are absent-minded. There are still c clear-headed people, and person k has moved up one position in line, so k should be replaced by $k - 1$. But since both n and k decrease by 1, j remains fixed. So the probability that (original) person k sits in the right seat will be $P(a - 1, c, j)$ in this case.

Combining these yields $P(a, c, j) = \frac{a+j}{n}P(a-1, c, j) + \frac{k-1-a}{n}P(a, c-1, j)$.

Now perform a strong double-induction on a and c . In the base cases where $a = 0$ or $c = j + 1$ the probability that person k was in the correct seat was $\frac{j+1}{j+a+1}$. So assume that $P(a, c, j) = \frac{j+1}{j+a+1}$ when either $a < A$ or $c < C$ for some A and C . The result is that

$$\begin{aligned}
P(A, C, j) &= \frac{A+j}{n} P(A-1, C, j) + \frac{k-1-A}{n} P(A, C-1, j) \\
&= \frac{A+j}{n} \frac{j+1}{j+A-1+1} + \frac{k-1-A}{n} \frac{j+1}{j+A+1} \\
&= \frac{j+1}{n} \left(1 + \frac{k-1-A}{j+A+1} \right) \\
&= \frac{j+1}{n} \frac{j+k}{j+A+1}
\end{aligned}$$

and noting that $n = j + k$ leaves $\frac{j+1}{j+A+1}$ as desired.

Problem 17. Let $c = n - a$. The proof will proceed by induction on c . Clearly when $c = 0$ there are $n = a$ people, all of whom are absent-minded, so all sit in seats chosen uniformly at random. Obviously there are $a!$ arrangements.

Claim: If $A = (p_1, p_2, \dots, p_n)$ is a possible arrangement of $n = a + c$ where the first a are absent-minded, there are exactly $a + 1$ possible arrangements of $n + 1$ people that can be obtained from it by adding one more person, p_{n+1} . These arrangements can be obtained from A by appending p_{n+1} to the end of A , or by placing p_{n+1} in one of the first a positions and moving the person currently in that position to the end of the arrangement. The $a + 1$ arrangements so obtained are all distinct from each other, and are distinct from any arrangement obtained by applying the same operation to any other possible arrangement of n people, and all possible arrangements are obtainable in this way.

The claim says that any possible arrangement can be built up one person at a time from exactly one possible arrangement of fewer people, and the building process can be done in exactly one way. According to the claim there are $a + 1$ arrangements of $a + c + 1$ people obtainable from arrangements of $a + c$ people. So the number of possible arrangements is multiplied by $a + 1$ each time a person is added. So using the $a!$ arrangements of the first a people, $c = 0$, as a base case, simple induction on c shows that the formula for the number of possible arrangements is correct if there are $n = a + c$ people in total.

For example, if $a = 3$, one possible arrangement of seven people would be $(7, 5, 2, 4, 3, 1, 6)$. This happened because persons 1, 2, and 3 came in and sat randomly in seats 6, 3, and 5 respectively. When person 4 came in, her seat was available so she took it. Then person 5's seat was unavailable, so she randomly sat in seat 2. Person 6 then randomly sat in seat 7, and person 7 had to take the only available seat, seat 1. According to the claim, the following arrangements are all possible for $a = 3$ with eight people: $(7, 5, 2, 4, 3, 1, 6, 8)$, $(8, 5, 2, 4, 3, 1, 6, 7)$, $(7, 8, 2, 4, 3, 1, 6, 5)$, and $(7, 5, 8, 4, 3, 1, 6, 2)$. Checking that these are possible arrangements is routine; clearly they are all distinct from each other. It is not quite as obvious that none of these could be obtained by starting with a different possible arrangement of seven people, nor is it obvious that *every* possible arrangement of eight people can be obtained by adding an eighth person to some possible arrangement of seven people in this way. Those are the contents of the claim.

There are several parts to the claim. Going through them methodically:

Let $A = (p_1, p_2, \dots, p_a, p_{a+1}, \dots, p_{a+c})$ be a possible seating arrangement of $n = a + c$ people. Then if there are $n + 1$ people, the first n can certainly sit in the seats shown by arrangement A as long as none of them randomly choose to sit in seat $n + 1$. In this case, person $n + 1$ sits in seat $n + 1$ and that is the first of the arrangements indicated in the claim. On the other hand, anyone sitting in seat s where $s \leq a$ chose to do so at random, because those seats belong to the absent minded-people, and the only time someone does not have a choice is if she is not absent-minded and her seat is available. So if the person sitting in seat s had instead chosen to sit in seat $n + 1$ and everyone between that person and person $n + 1$ makes the same choice as they did in forming arrangement A , then seat s will be the only seat available to person $n + 1$, so they will perforce sit there, obtaining the other arrangements indicated by the claim. So each arrangement of n people leads to $a + 1$ arrangements of $n + 1$ people.

These $a + 1$ arrangements are clearly distinct from each other, because person $n + 1$ is in a different seat in each one.

If A' is an arrangement of $n + 1$ people obtained from arrangement A , and B' is obtained from arrangement B , then A' and B' can only be equal if A and B were the same to start with. This is because for A' and B' to be equal, person $n + 1$ must be in the same seat in both. If this is the last seat in both, then A and B were originally just $A' = B'$ without the last entry. If, instead, this is seat s with $1 \leq s \leq a$ then A' and B' can only be equal if they have the same person sitting in the last chair. But then A is just A' with this person moved from chair $n + 1$ to chair s and then the last chair being taken away. The same is true of B . All other people in these arrangements must be in the same chairs in both A and B , so A and B must be equal. Thus, arrangements coming from different arrangements of n people lead to different arrangements of $n + 1$ people.

Finally, it must be shown that all possible arrangements of $n + 1$ people can be obtained from some arrangement of n people by the method of the claim. So let A' be a possible arrangement of $n + 1$ people. By problem 15, person $n + 1$ can only be in seat s where $1 \leq s \leq a$ or $s = n + 1$. Clearly if $s = n + 1$ then all of the first n people chose or were required to sit in the first n seats, so A obtained from A' by deleting the last entry (representing that person $s + 1$ is sitting in seat $s + 1$) is a possible arrangement of n people, and one that can be expanded as in the claim to obtain A' .

On the other hand, if $1 \leq s \leq a$ let the person who is sitting in seat $n + 1$ be person p . It must be shown that the arrangement A obtained from A' by substituting p instead of p_{n+1} in seat s and eliminating seat $n + 1$ is a possible seating arrangement of n people. But this is certainly the case, because seat s was available to person $n + 1$, so it would certainly have been available earlier to person p when they chose to sit in seat $n + 1$. If they had chosen to sit in seat s , and everyone else makes the same choices as they made to form A' , then just before person $n + 1$ comes into the room the first n people are sitting in arrangement A .

The author does not know of any practical applications of this problem. He just thought it was a curious result. A recent article on this problem in the The American Mathematical Monthly uses statistical methods and random variables to get at some of these questions in more depth. The article is: Norbert Henze and Günter Last, *Absent-Minded Passengers*, The American Mathematical Monthly, 126.10, December 2019. The article has references to other sources on the problem and its extensions.

One possible extension is: what if the absent-minded people are not grouped together at the beginning of the group? It should not be too hard to show that for any given clear-headed person, all that matters is the number of absent-minded people who preceded her into the room, not whether they were all grouped together or not.

And in case you were wondering if there was any relation between the counting ideas related to this problem, where an $(a + 1)^{n-a}$ appears, and the parking sequences problem from last year, where an $(a + 1)^{a-1}$ appears, well, I think you already know the answer to that ...